Ladder Qubits

Analysis of a Spin Ladder with Quantum Error Correcting and Detecting Properties



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Nature's Computers

The Computer Science Assumption: Discrete storage and discrete manipulation of information can be robustly enacted.

The theory of classical fault tolerance guarantees this assumption for classical information, but in practice, there are **DISTINCT PHYSICAL REASONS FOR THE STABILITY OF CLASSICAL INFORMATION.** i.e. there are physical systems which are naturally fault tolerant classical computers.

similarly

The theory of quantum fault tolerance guarantees that the computer science assumption works for quantum information.

Are there physical systems which are naturally fault tolerant quantum computers?

Oh Brave New Qubits!

Goal: proof of principle of natural fault tolerant quantum computation

What this really means: Are there *physically realistic* ***Hamiltonians*** for a *many-body system* which exhibit

A. Eigenstates of the Hamiltonian are quantum error correcting code states. (error correction)

B. Persistent coherence. Existence of an order parameter for coherence which is nonvanishing at nonzero temperature. (natural error correction)

C. Discrete set of manipulations of this order parameter which is form a universal set of gates and whose faulty implementation are errors (natural fault tolerance)

which is not to suggest we should show a Caliban-like (John the Savage-like?) despair towards the present theory of fault tolerant quantum computation!

Prior Inspirations

Kitaev's toric codes: First example of a quantum error correction code for a many-body Hamiltonian (A).

Unfortunately toric codes use greater than two body interactions and to make universal these codes require many level systems coupled in an unnatural manner. Also coherence order parameter vanishes exponentially for nonzero temperature (supplement by error correction to make fault tolerant)

Barnes and Warren's NMR code: Uses only two body interactions in a realistic NMR setting, BUT is only a classical error correcting code. (not different from 1D Ising model) Notion of a "funnel" of energy levels to make error correction natural is identically the concept of persistent coherence order parameter.

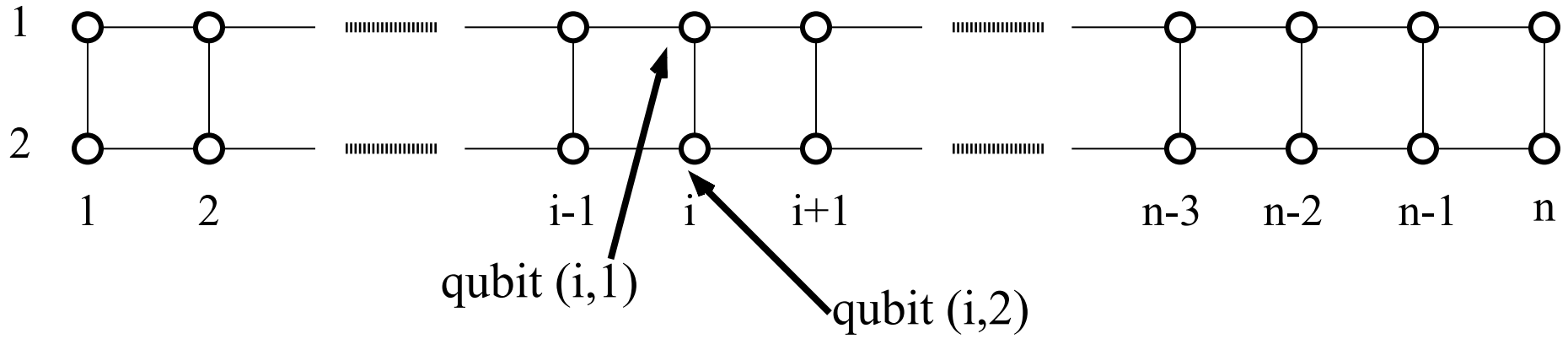
Freedman's quantum loop gas: Evidence, but no concrete realization, for naturally fault tolerant quantum computer.

see also work of Preskill

Summary of Results

- Spin ladder with 2n qubits and a Hamiltonian with two qubit interactions along rungs and legs.
- Ground state is two-fold degenerate: can be used to store qubit.
- Ground state is quantum error correcting code for $\left\lfloor \frac{n-1}{2} \right\rfloor$ phase errors and quantum error detecting code for single bit flip errors
- Exact calculation of the spectrum of the spin ladder shows gap for phase and bit flip excitations
- Generalization of spin ladder to two dimensional spin lattice yields the a ground state which is fully error correcting
- Baby step towards naturally fault tolerant quantum computation

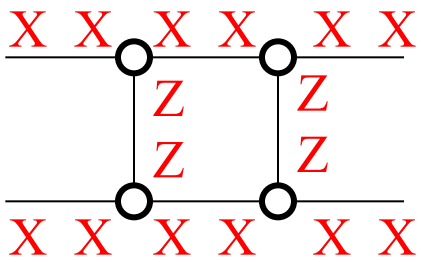
Description of the Spin Ladder



$\sigma_\alpha^{(i,j)} \equiv \alpha^{\text{th}}$ Pauli operator acting on the (i, j) qubit tensored with I on all other qubits

The Hamiltonian

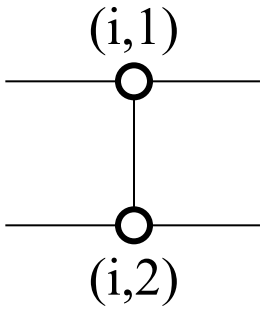
$$H = -\omega \left[\sum_{i=1}^{n} \sigma_Z^{(i,1)} \sigma_Z^{(i,2)} + \sum_{i=1}^{n-1} \left(\sigma_X^{(i,1)} \sigma_X^{(i+1,1)} + \sigma_X^{(i,2)} \sigma_X^{(i+1,2)} \right) \right]$$



Twisted Ladder Basis

For rung i of the ladder define

$$X_1^{(i)} = \sigma_X^{(i,1)}\sigma_X^{(i,2)}, \quad Z_1^{(i)} = \sigma_Z^{(i,1)}$$
$$X_2^{(i)} = \sigma_X^{(i,2)}, \quad Z_2^{(i)} = \sigma_Z^{(i,1)}\sigma_Z^{(i,2)}$$



Notice $X_k^{(i)}, Z_k^{(i)}$ are simply two qubit Pauli operators in a different basis (related to old by controlled-not gate). We will use the basis specified by these operators via

$$Z_2^{(i)}\left|z_2^{(1)}, z_2^{(2)}, \ldots, z_2^{(n)}, x_1^{(1)}, x_1^{(2)}, \ldots, x_1^{(n)}\right\rangle = z_2^{(i)}\left|z_2^{(1)}, z_2^{(2)}, \ldots, z_2^{(n)}, x_1^{(1)}, x_1^{(2)}, \ldots, x_1^{(n)}\right\rangle$$

$$X_1^{(i)}\left|z_2^{(1)}, z_2^{(2)}, \ldots, z_2^{(n)}, x_1^{(1)}, x_1^{(2)}, \ldots, x_1^{(n)}\right\rangle = x_1^{(i)}\left|z_2^{(1)}, z_2^{(2)}, \ldots, z_2^{(n)}, x_1^{(1)}, x_1^{(2)}, \ldots, x_1^{(n)}\right\rangle$$

where $z_2^{(i)}, x_1^{(i)} \in \{\pm 1\}$

To fit on the poster(!), use the shorthand

$$\left|z_2, x_1\right\rangle \equiv \left|z_2^{(1)}, z_2^{(2)}, \ldots, z_2^{(n)}, x_1^{(1)}, x_1^{(2)}, \ldots, x_1^{(n)}\right\rangle$$

which we will call, the “twisted ladder basis”

Ising Coefficients

The ladder Hamiltonian is, in terms of the twisted ladder basis operators

$$H = -\omega\left[\sum_{i=1}^{n} Z_2^{(i)} + \sum_{i=1}^{n-1} X_2^{(i)} X_2^{(i+1)}\left(I + X_1^{(i)} X_1^{(i+1)}\right)\right]$$

Sinc
e

$$\left(I + X_1^{(i)} X_1^{(i+1)}\right)\left|z_2, x_1\right\rangle = \left(1 + x_1^{(i)} x_1^{(i+1)}\right)\left|z_2, x_1\right\rangle$$

$$\left(1 + x_1^{(i)} x_1^{(i+1)}\right) = \begin{cases} 0 \text{ if } x_1^{(i)} \neq x_1^{(i+1)} \\ 2 \text{ if } x_1^{(i)} = x_1^{(i+1)} \end{cases}$$

We can write the ladder Hamiltonian as

$$H = -\omega\left(\bigoplus_{x_1^{(1)}, x_1^{(2)}, \ldots, x_1^{(n)} \in \{\pm 1\}} \left[\sum_{i=1}^{n} Z_2^{(i)} + 2\sum_{i=1}^{n-1} c_i\left(x_1\right) X_2^{(i)} X_2^{(i+1)}\right] \otimes \left|x_1\right\rangle\left\langle x_1\right|\right)$$

$$c_i\left(x_1\right) = \frac{1}{2}\left(1 + x_1^{(i)} x_1^{(i+1)}\right) \in \{0, 1\}$$

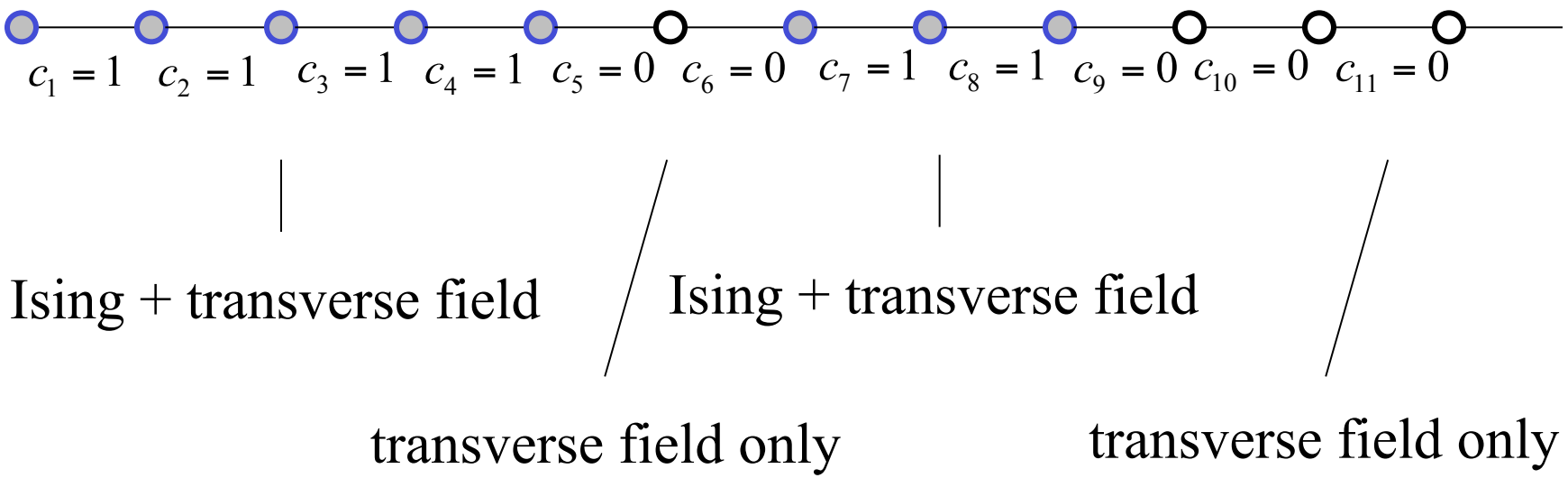
This is block diagonal where each block corresponds to a given $\left|x_1'\right\rangle$
Given x_1' the form of this block follows from the Ising coefficients $c_i\left(x_1\right)$

Clusters, Clusters, Everywhere

For a given $\underline{x}_1$ the block Hamiltonian is

$$\sum_{i=1}^{n} Z_2^{(i)} + 2\sum_{i=1}^{n-1} c_i(x_1) X_2^{(i)} X_2^{(i+1)}, \qquad c_i(x_1) \in \{0,1\}$$

Which is the Hamiltonian for clusters of "Ising chains in a transverse field" and "transverse field only"



Note that a cluster of "Ising+transverse" need not be separated from another cluster of "Ising+transverse" by "transverse only"

Jordan, Wigner, Bogoliubov

Length k spin chain of an Ising field with a transverse field can be exactly solved using a Jordan-Wigner transformation and a generalized Bogoliubov transformation:

Jordan-Wigner transformation: maps spins in a linear array to spinless fermions.

Bogoliubov transformation: diagonalizes quadratic fermion Hamiltonain.

$$H_k = -\omega\left(\sum_{i=1}^{k} Z_2^{(i)} + 2\sum_{i=1}^{k-1} X_2^{(i)} X_2^{(i+1)}\right) = 2\sum_q \sqrt{5+4\cos(q)}\left(\eta_q^\dagger \eta_q - \frac{1}{2}\right) + \Omega_k, \quad \mathrm{Tr}[\Omega_k] \le \frac{4}{k}$$

$\eta_q^\dagger, \eta_q$ are fermion operators corresponding to some combination of the $X_2^{(i)}, Z_2^{(i)}, iX_2^{(i)}Z_2^{(i)}$ operators

$\sum_q$ runs from $\frac{2\pi q}{k} = -\frac{k}{2}\left(-\frac{k-1}{2}\right)$ to $\frac{2\pi q}{k} = \frac{k}{2}\left(\frac{k-1}{2}\right)$ for k even(odd)

Cluster Structure Energy

Energy of the ground state (vacuum) for a cluster of length k is

$$E_g(k) = -\omega \sum_{m=-\frac{k}{2}\left(-\frac{k-1}{2}\right)}^{\frac{k}{2}\left(\frac{k-1}{2}\right)} \sqrt{5 + 4\cos\left(\frac{2\pi m}{k}\right)} + O\left(\frac{\omega}{k}\right)$$

Energy for the ground state of a given block Hamiltonian specified by $\underline{x}_1$ depends only on the number and size of clusters

$$E\left(\left[m_2, m_3, \ldots, m_{n-1}\right]\right) = \sum_{i=2}^{n-1} m_i E_g(i) - \omega\left(n - \sum_{i=2}^{n-1} m_i\right)$$

$m_i \equiv$ number of "Ising+transverse" clusters of size i

Ground state corresponds to the vacuum state for the full cluster configuration $\left[m_2, m_3, \ldots, m_{n-1}\right] = [0, 0, \ldots, 0, 1]$ or $c_1 = c_2 = \left| \right. = c_{n-1} = 1$

Excitations within clusters (fermion creation) cost $\approx 2\omega$ and those which move between full cluster and other clusters cost $\approx \omega$

All a Bunch of Degenerates

Recall $c_i(x_1) = \frac{1}{2}\left(1 + x_1^{(i)} x_1^{(i+1)}\right)$ so there is a two-fold degeneracy in the assignment of a given *ci* via a specification of the $x_1^{(i)}$ (flipping sign of every $x_1^{(i)}$ does not change any of the *ci*) This degeneracy corresponds to a two-fold degeneracy of every energy level.

The algebra generated by the rungs and legs of the spin ladder, i.e. the algebra generated by the set

$$\left\{\sigma_Z^{(i,1)}\sigma_Z^{(i,2)}, \forall 1 \le i \le n\right\}\ldots\left\{\sigma_X^{(i,1)}\sigma_X^{(i+1,1)}, \sigma_X^{(i,2)}\sigma_X^{(i+1,2)}, \forall 1 \le i \le n-1\right\}$$

has a nontrivial commutant (operators which commute with everything in the algebra) generated by operators

$$\prod_{i=1}^{n}\sigma_Z^{(i,1)} \text{ and } \sigma_X^{(1,1)}\sigma_X^{(1,2)}$$

These operators act on the above degeneracy as the encoded Pauli operators.

When is a Ladder a Quantum Error Correcting Code?

Errors on the lattice which are products of single bit flip errors (σX) and *n-1* phase errors (σZ) modulo the algebra of the spin ladder are not logical operators on the code. Hence

The ground state of the spin ladder is

an error detecting code for single bit flip errors

an error correcting code for up to $\left\lfloor \frac{n-1}{2} \right\rfloor$ phase flip errors

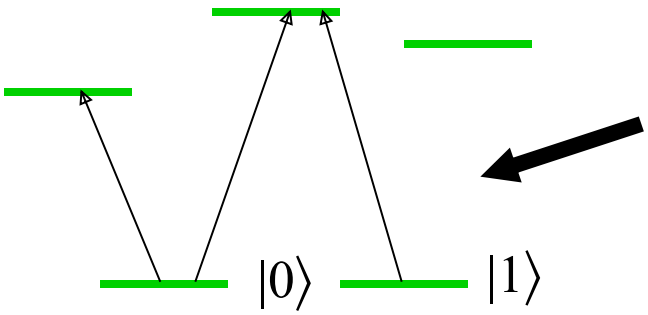
For our "Brave New Qubits", we have partially satisfied the condition

A. Eigenstates of the Hamiltonian are quantum error correcting code states. (error correction)

The Gap is Repressive

Fight the Gap

Furthermore, there is a gap for excitations when the first of either such error occur. This leads to a repressive (but NOT to naturally error correction) of these errors at low temperatures.



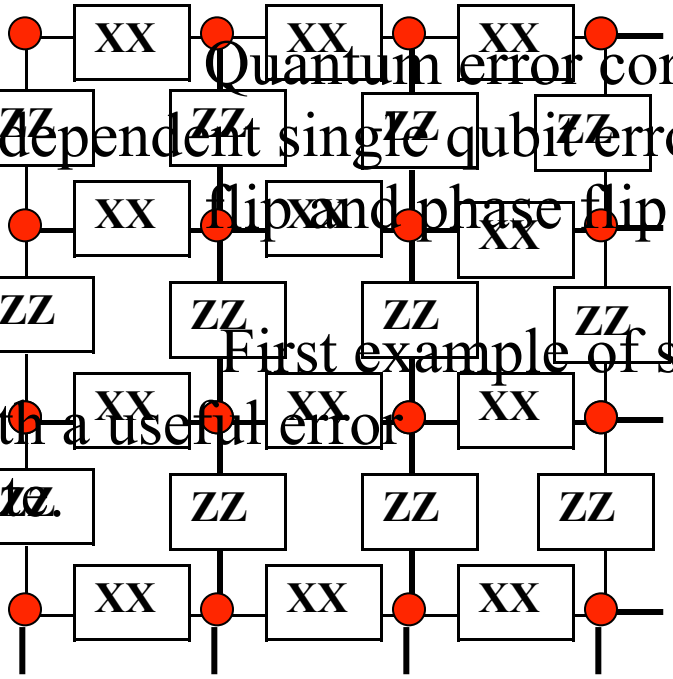
All single qubit errors are energetically unfavorable
= supercoherence.

Because the ground state is also quantum error correcting for phase errors, one can also perform error correction for these phase errors to supplement the supercoherent effect. (Helping the Gap can make it more repressive)

The correction corresponds to measuring and correcting the eigenvalues corresponding to the *ci*

The Lattice Generalization

Obvious generalization to lattice system has nicer properties:



Quantum error correcting for independent single qubit errors (bit flip and phase flip and combo)

First example of such a two qubit interaction with a useful error correcting code as its ground state.

Problems(?) for this lattice: (1) unknown whether there are gapless excitations, (2) still no natural error correction (like 1D Ising model, does not naturally correct errors and unlike 2D Ising model with does naturally correct errors), (3) Universal set of gates on code seems difficult.

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Professor K. Brigitta Whaley

Department of Chemistry, University of California, Berkeley

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